

Toward a Next Generation Agricultural Expert System

A Multimodal Conversational Approach

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Introduction: The Agricultural Advisory Crisis in India

- **Extension Shortage:** Only 1 extension worker per 1,000+ farmers in India — 146 million farm households make critical crop decisions with zero professional guidance [6].
- **Crop Loss:** 26–30% of annual crop production is lost to disease and pests, yet most farmers identify symptoms by eye — well past the optimal treatment window [2].
- **Data Inaction:** 230 million Soil Health Cards issued but only 14.6% of farmers acted on them — no system translates soil data into actionable advice in their own language [11].
- **Price Gap:** Farmers sell 30–40% below optimal price due to lack of real-time mandi data and price forecasting — a gap affecting over 120 million households [1].
- **Access Barriers:** Existing agricultural AI is English-only, text-only, and connectivity-dependent — structurally inaccessible to the majority of Indian farmers it claims to serve.

Related Work & Research Gaps

System	Venue	Multi-agent Router	VLM Disease	Soil Sensor	Weather Prediction	XAI Farmer UI	Voice / Indic	Market Price	Deployed
Farmer.Chat [16]	ACM 2024	×	× (Generic)	×	✓ API	×	✓ Partial	×	✓ 565k users
AgroAskAI [4]	arXiv 2024	✓ Chain	×	×	✓ Dedicated	×	✓ Partial	×	×
AgriGPT [18]	arXiv 2025	✓ Data engine	✓ VL ext.	×	✓ Partial	×	×	×	×
AgriM-LLM [17]	Preprints 2026	×	✓ Best VE	×	×	×	×	×	×
AgriDoctor (Base) [9]	IEEE 2025	✓ Router+5	✓ Strong	×	×	×	✓ CN/EN only	×	×
AgriBuddy [12]	BRAC 2025	✓ Partial	✓ CNN rice	×	✓ Partial	×	×	×	✓ Partial
PhenoAssistant [15]	Nature 2026	✓ Yes	✓ Research	×	×	×	×	×	× (Lab)
Agentic AI [7]	MDPI 2026	✓ 4 agents	✓ Rice CNN	✓ Partial	✓ Dedicated	×	×	×	×

Key Research Gaps Identified

- **Soil Sensor Fusion** and **Market Price** features are completely missing from existing systems.
- No prior platform supports explainable AI (**XAI**) for farmer UIs or comprehensive **Indic speech** interfaces.

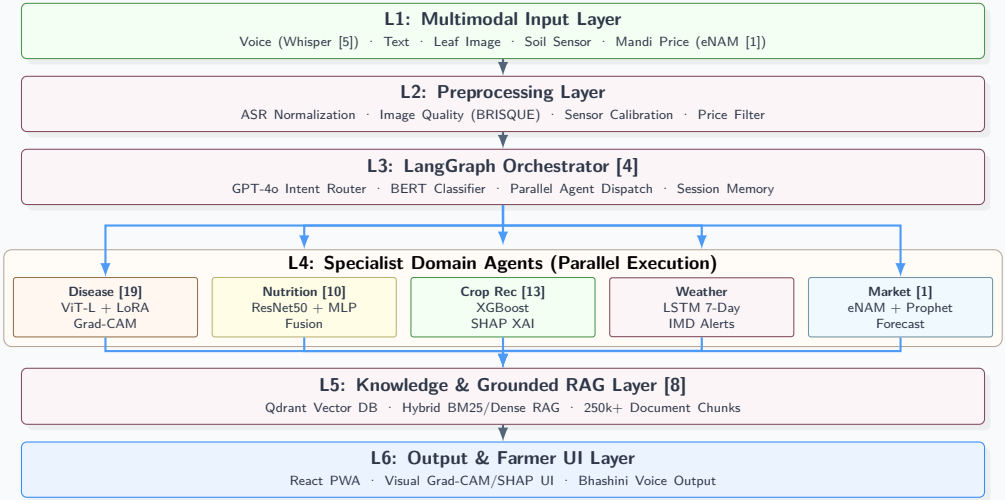
Problem Statement

Current agricultural AI systems are fatally fragmented: deployed chatbots lack visual diagnostics, advanced vision models lack interactive conversational interfaces, multi-agent setups ignore live market data, and none deliver explainable advice in native Indic languages. Consequently, no existing framework integrates speech, image, soil sensors, weather prediction and live market data into a single trustworthy session.

Research Objectives

- **O1: Multi-Agent Orchestration** — Design a LangGraph orchestrator to classify farmer query intent and route requests to domain-specific specialist agents.
- **O2: Disease & Nutrient Diagnostics** — Fine-tune a ViT-L/16 (LoRA) model for disease identification ($> 90\%$ accuracy) and fuse leaf image features with soil sensor telemetry.
- **O3: Grounded RAG & Explainable AI** — Build a BM25/dense vector RAG pipeline over 250k+ documents with LLM confidence gating, surfacing Grad-CAM and SHAP explanations.
- **O4: Speech, Weather & Market Advisory** — Implement a 10-language Indic Whisper/Bhashini voice pipeline; deploy an LSTM-based 7-day weather forecasting module using IMD data; and integrate live eNAM prices with Prophet-based mandi price forecasts.
- **O5: Agronomic Benchmarking & Validation** — Evaluate system performance against MIRAGE [14]/AgMMU [3] benchmarks and validate real farmer queries with 10 ICAR agronomists.

System Architecture



Research Contributions

- **C1: Multi-Agent System** — First deployment-ready multi-agent agricultural expert system spanning five specialist domains.
- **C2: Cross-Modal Fusion** — Cross-modal fusion of leaf image and soil sensor data for nutrition deficiency advisory in a conversational loop.
- **C3: Farmer-Facing XAI** — Farmer-facing explainable AI integrated into a conversational web interface.
- **C4: Indic Voice Pipeline** — End-to-end Indic voice advisory pipeline connecting ASR, multilingual RAG, and TTS.
- **C5: Mandi Price Advisory** — Live mandi price advisory co-delivered with agronomic advice in one conversational turn.

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Thank You!

Questions & Discussion

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